

## 12 M001 Week 3 — English Worksheet (Beginner)

**Topic:** Variables · **Course:** Pyramid Problems · **Level:** Beginner · **Time:** about 30 minutes

### 1 · Predict 🧠

Read the steps. Before you imagine the agent doing it, circle or write your answer.

```
set size to 3
repeat size times:
  place block down
  move forward
```

**How many blocks does the agent place? Circle one:** 1 · 3 · 6

**Where does the number 3 come from?**

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```
set size to 6
repeat size times:
  place block down
  move forward
```

**Same code, new number. How many blocks now? Circle one:** 3 · 6 · 12

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## 2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

**Pair A** — The agent should place a row of `size` blocks, where size is 5.

```
# clean
set size to 5
repeat size times:
  place block down
  move forward
```

```
# buggy
set size to 5
repeat 3 times:
  place block down
  move forward
```

What is wrong? Which number does the loop really use?

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**Pair B** — The agent should build a **tower** that is `height` blocks tall.

```
# clean
set height to 4
repeat height times:
  place block down
  move up by 1
```

```
# buggy
set height to 4
repeat height times:
  place block down
  move forward
```

What is wrong?

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### 3 · Fill the Gap

The agent should build a row of 4 blocks using a variable. One line is missing. Fill it in using the word bank.

```
_____  
repeat size times:  
  place block down  
  move forward
```

**Word bank:** set size to 4 · move forward · repeat 4 times

**Write the missing line:**

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## 4 · Tell Me What You Built 📷

Now switch to your homework world. Use a `size` variable to build a **size-3 wall**. Then change the variable to **6** and run it again. When you finish, come back here.

Send a photo or video of both walls, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

I made a variable called ...

When I changed it to 6, the wall ...

A variable is useful because ...

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## 5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) and show your walls. Talk like you are teaching a friend. Try to use these words: **variable**, **set**, **loop**, **repeat**.

1. Show the small wall and the big wall.
2. Read your variable line out loud.
3. Change the number one more time and show what happens.

**Write what you will say in your video.** You can read from it while filming.

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**Submit** ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.